ARNAV SHARMA

+91-9039415506 • arnavsharma090406@gmail.com • github.com/arnavsharma09 • linkedin.com/in/arnav-sharma09 • arnavsharma09.github.io • Delhi, India

SUMMARY

Mechanical engineering student at IIT Jodhpur building a career in data science and machine learning. Approaches ML the way an engineer approaches a structure: find how it fails, measure how much it can take, and be honest about where its limits are. Cares more about results that hold up under scrutiny than results that look impressive, and wants to work on systems people can trust. Seeking a Data Science, ML, or AI Engineering internship.

SKILLS

Languages: Python, SQL, C, C++

Machine Learning: Scikit-learn, Regression (Ridge, Lasso), Logistic Regression, KNN, SVM, Naive Bayes, Decision Trees, Random Forest, XGBoost, Ensemble Methods, K-Means, DBSCAN, PCA, t-SNE, Anomaly Detection, Cross-Validation, Optuna, SHAP, SMOTE, Multi-Armed Bandits (UCB1)

Deep Learning: PyTorch, TensorFlow, Keras, CNNs, RNNs, LSTMs, Transformers, Transfer Learning, Embeddings, GANs, VAEs

LLM / Multi-Agent Systems: LangGraph, LangChain, Anthropic Claude API, ChromaDB, RAG, DeepEval, Agentic AI Workflows, Prompt Engineering, LLM Security / Red-Teaming, Prompt-Injection Defense, Guardrail Design, LLM Evaluation, Model Context Protocol (MCP)

Data Engineering, Analytics & BI: PostgreSQL, SQL (CTEs, window functions, dimensional modeling), Pandas, NumPy, Polars, Feature Engineering, Statistical Hypothesis Testing (scipy, statsmodels), Confidence Intervals, EDA, Power BI, Tableau, Matplotlib, Seaborn, Plotly, ETL, Data Cleaning

Tools & Platforms: Git, Docker, REST APIs, FastAPI, Streamlit, Ollama, pytest (unit testing), MLflow, Linux, Jupyter, Google Colab

PROJECTS

Agent Red-Team | Python, Ollama (Qwen2.5-7B), UCB1 Bandit, Wilson CIs, pytest

- Built an adaptive red-team harness for tool-calling LLM agents: a UCB1 bandit searches 144 prompt-injection recipes against a local Qwen2.5-7B agent; undefended, injections succeeded in 17–37% of episodes (37.0% destructive, 17.4% exfiltration over 350 episodes).
- Implemented an argument-provenance taint guardrail: exfiltration succeeded in 0/250 episodes vs 61/350 undefended, with 0/36 benign-task blocks (Wilson 95% upper bound 9.6%) and 35/36 task completion.
- Found and fixed a gap in my own defense (file reads trusted: 15/31 exec episodes got through), then closed a base64 self-decoding bypass (0/100 executions after); 191 unit tests, limits documented.

AutoInsight | LangGraph, Polars, ChromaDB, FastAPI, Streamlit, Anthropic Claude API, Docker

- Architected a 7-node multi-agent LangGraph pipeline (Profiler, Router, Planner, Coder, Sandbox, Critic, Reporter) that plans and executes 3–6 step statistical analyses using lazy-evaluated Polars pipelines, with a human-in-the-loop approval checkpoint before any code executes.
- Built a hybrid methodology router (rule-based matching + ChromaDB semantic search across 60+ statistical test descriptions) and a Critic agent that validates methodology against Shapiro-Wilk normality results, auto-correcting to non-parametric equivalents (e.g. ANOVA → Kruskal-Wallis + Dunn post-hoc) with bounded retry logic (max 3 retries/step); covered core logic with an 18-test automated suite at 100% pass rate.
- Sandboxed all LLM-generated code execution in an isolated Docker container with AST-level security scanning (blocking os, subprocess, eval, and sandbox-escape patterns), and validated report faithfulness against computed metrics using DeepEval to catch hallucinated numbers.

RetainIQ | PostgreSQL, SQL, Python (scipy, statsmodels), Power BI, Docker

- Built an end-to-end analytics pipeline on 99,441 orders across 9 relational tables (Brazilian e-commerce dataset), modeling a star schema in PostgreSQL and applying quintile-scored RFM segmentation across 94,983 customers into 7 value tiers.
- Ran a controlled hypothesis test isolating delivery time's effect on customer satisfaction independent of distance (Haversine-computed): Shapiro-Wilk confirmed non-normality, so Mann-Whitney U was used as the pre-committed primary test, finding fast-delivery orders scored significantly higher (4.31 vs 3.75, $p<0.001$).
- Identified a 3.04% repeat-purchase rate and 62.5% revenue concentration across just 3 of 27 states; shipped a 6-visual interactive Power BI dashboard (KPI cards, cohort retention heatmap, RFM breakdown, geographic delivery map) translating findings into 5 stakeholder-ready recommendations.

EDUCATION

Indian Institute of Technology, Jodhpur Aug 2024 – May 2028 (Expected)
B.Tech, Mechanical Engineering

COURSEWORK & CERTIFICATIONS

Relevant Courses: Machine Learning, Data Structures & Algorithms, Linear Algebra, Probability & Statistics

Certification: Data Science Master Program – CampusX | Topics: ML, Deep Learning, SQL, EDA, Feature Engineering

LEADERSHIP & POSITIONS OF RESPONSIBILITY

Assistant Head, Marketing – Prometeo'26 | IIT Jodhpur 2025 – 2026

- Owned end-to-end marketing strategy for IIT Jodhpur's flagship annual technical festival; led a cross-functional team across outreach, sponsorship, and digital channels to maximize audience reach and partner acquisition.

Student Guide – Student Wellbeing Committee | IIT Jodhpur 2025 – 2026

- Guided a cohort of incoming first-year students through academic transitions and personal challenges; served as a primary peer-support contact within the institute's structured student welfare framework.